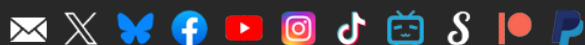


Inigo Quilez :: articles :: trigonometric functions



## Intro

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These are some of the functions involving trigonometrics that I keep encountering over and over again while I do computer graphic explorations. Many of them aren't listed in [Wikipedia](#), so I end up deriving them again each time. So this is the list, mostly for myself, where I collect the results so I save myself, and maybe you too, some time next time they appear in your work.

## Adding waves

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These are related to QAM modulation, and as an electrical engineer I'm familiar with them, but I leave them here since they appear often for me when finding minima and maxima of different SDF shapes.

$$a \cos(x) + b \sin(x) = \sqrt{a^2 + b^2} \cos\left(x - \arctan \frac{b}{a}\right)$$

$$a \cos^2(x) + b \sin^2(x) = \frac{a+b}{2} + \frac{a-b}{2} \cos(2x)$$

## Half angle trigonometric vs radicals

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These appear often when a quadratic quantity is lurking around. They are related to the [trisect\(\) function](#) at the core of cubic solvers, but one degree less.

$$\sin\left(\frac{\arcsin(x)}{2}\right) = \sqrt{\frac{1 - \sqrt{1 - x^2}}{2}} \cdot \text{sign}(x)$$

$$\cos\left(\frac{\arcsin(x)}{2}\right) = \sqrt{\frac{1 + \sqrt{1 - x^2}}{2}}$$

$$\sin\left(\frac{\arctan(x)}{2}\right) = \sqrt{\frac{1 - 1/\sqrt{1 + x^2}}{2}}$$

$$\cos\left(\frac{\arctan(x)}{2}\right) = \sqrt{\frac{1 + 1/\sqrt{1 + x^2}}{2}}$$

$$\sin\left(\frac{\arccos(x)}{2}\right) = \sqrt{\frac{1 - x}{2}}$$

$$\cos\left(\frac{\arccos(x)}{2}\right) = \sqrt{\frac{1 + x}{2}}$$

## Multiple angle trigonometric vs radicals

These showed up for me when working with the Mandelbrot set. They are kind of Chebyshev polynomials really...

$$\cos(2 \arccos(x)) = 2x^2 - 1$$

$$\sin(2 \arccos(x)) = 2x\sqrt{1-x^2}$$

$$\cos(4 \arccos(x)) = 8x^2(x^2 - 1) + 1$$

$$\sin(4 \arccos(x)) = 4x(2x^2 - 1)\sqrt{1-x^2}$$

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